

Role of ultrasound in evaluation of Neck LAP in children

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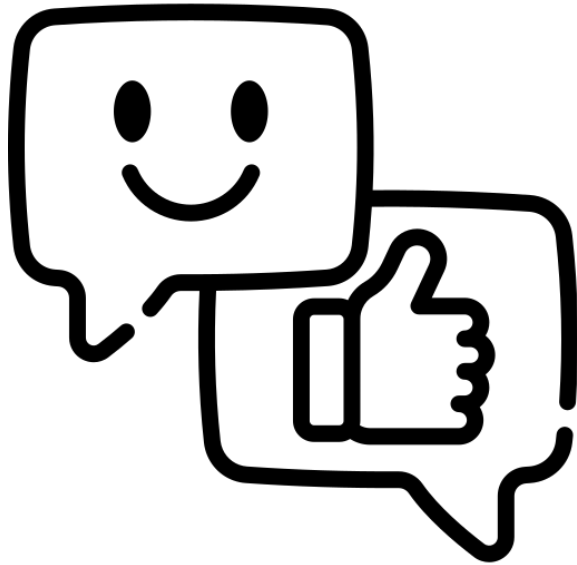
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Role of ultrasound in evaluation of Neck LAP

- Ultrasound is advocated for use as the initial imaging modality of choice for palpable neck masses in children to avoid radiation exposure from CT.
- It is useful in determining whether the lesion is solid or cystic in composition.
- It can be used to guide biopsies of potentially neoplastic masses or drainage of infectious fluid collections.

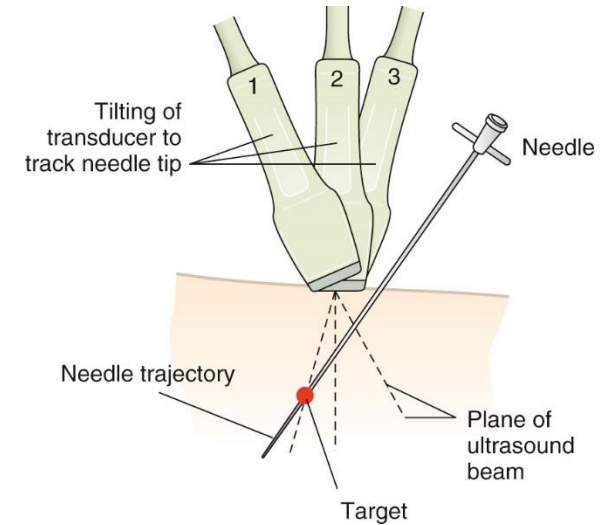
Ultrasound advantages



No
Radiation/Sedation

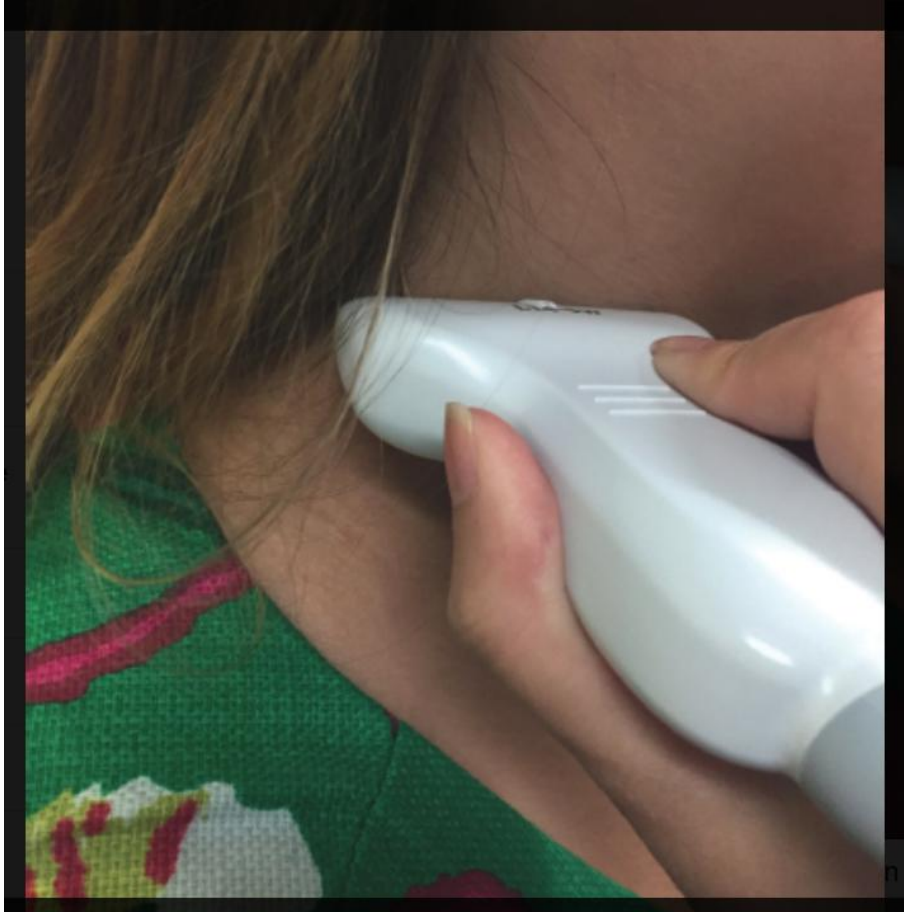


Portable
Real-time
Accessible



Guided-
Intervention

Standard Scan



Normal lymph node

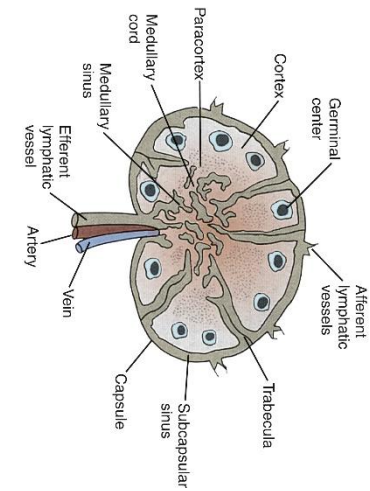


Fig. 1—7-year-old boy with normal lymph node. Gray-scale ultrasound shows circumscribed, ovoid node with cortical hypoechogenicity (*arrowhead*) and relatively hyperechoic hilum (*arrow*).

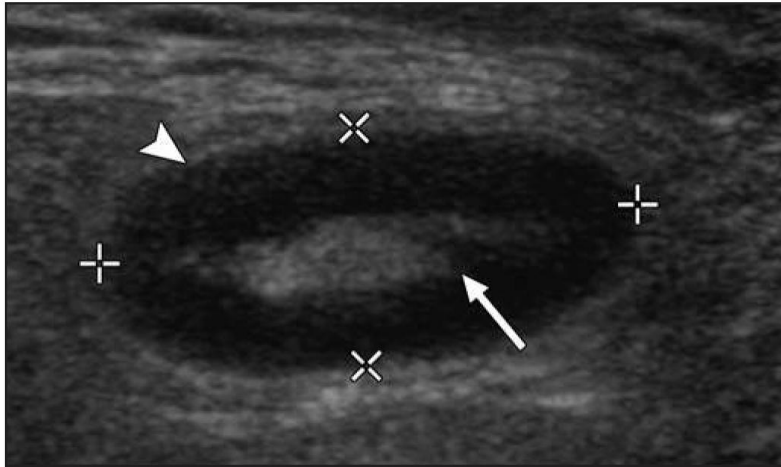
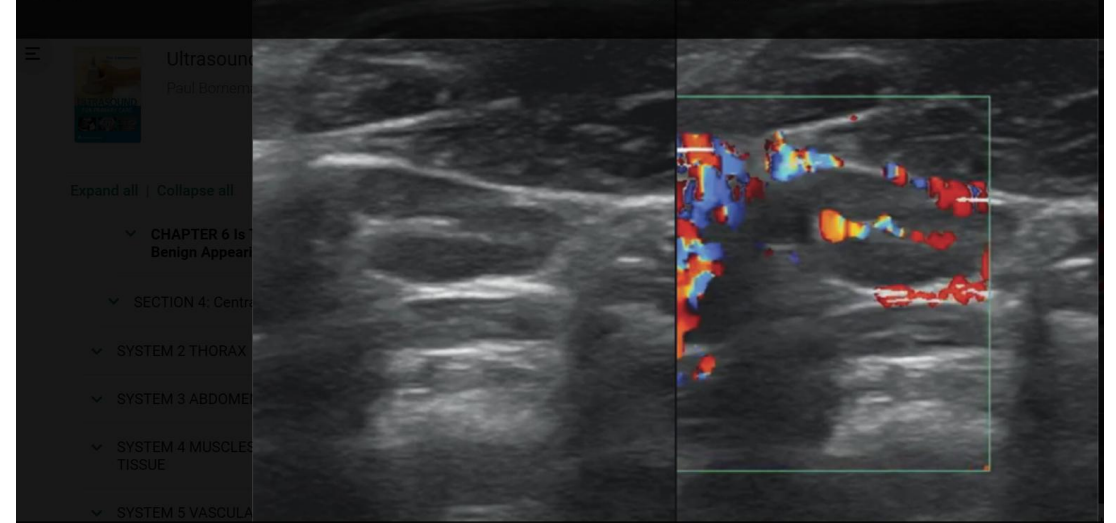
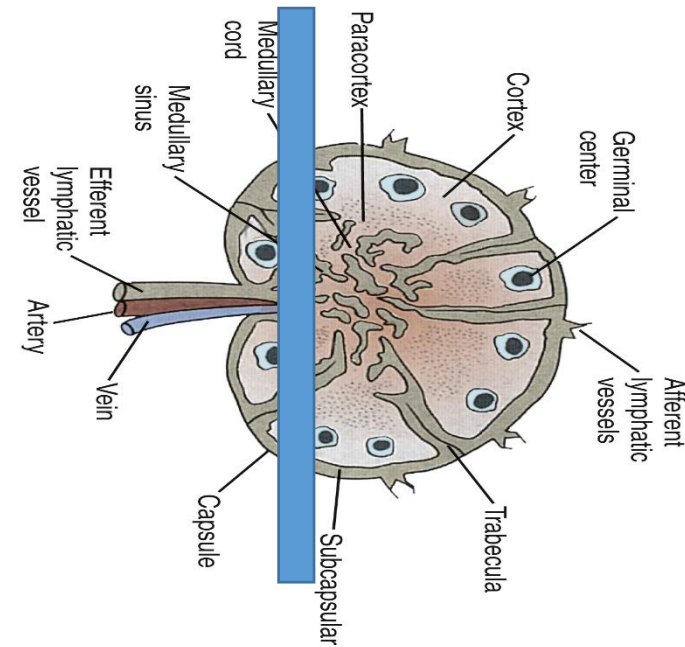
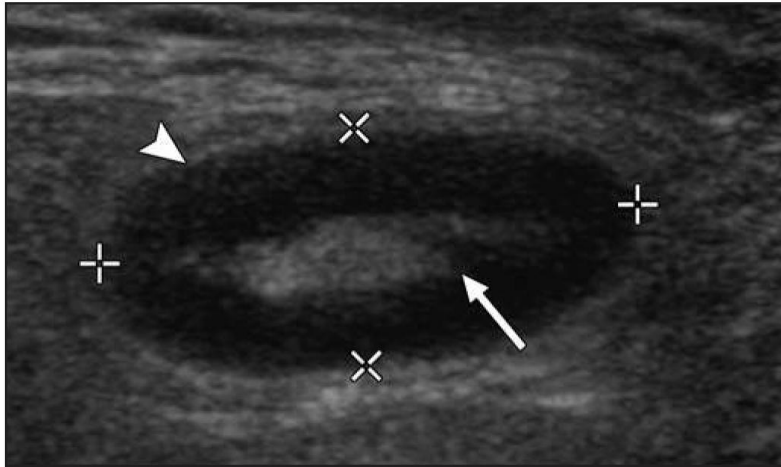


FIGURE 6-1. The normal oval lymph node has normal architecture (hilum and cortex). Color Doppler reveals central hilar flow.



Normal lymph node

Fig. 1—7-year-old boy with normal lymph node. Gray-scale ultrasound shows circumscribed, ovoid node with cortical hypoechogenicity (*arrowhead*) and relatively hyperechoic hilum (*arrow*).



Infected/Suppurative Lymph node

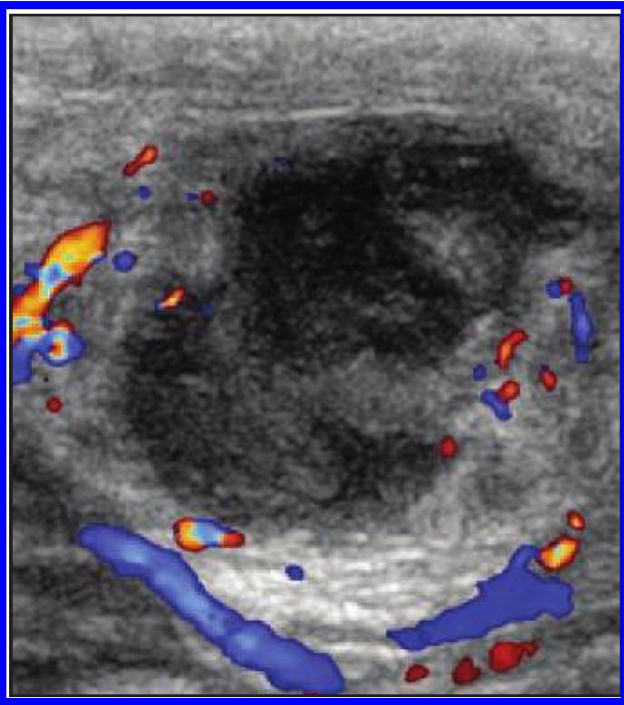
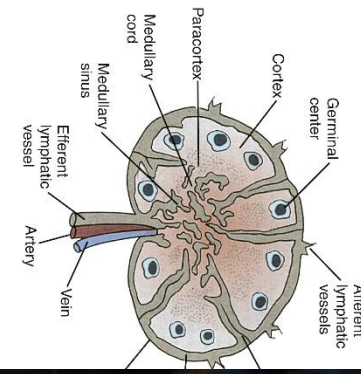


Fig. 5—6-year-old boy with suppurative lymph node due to *Staphylococcus aureus*. Gray-scale color Doppler ultrasound shows enlarged cervical lymph node with central hypoechoogenicity and peripheral Doppler vascularity.

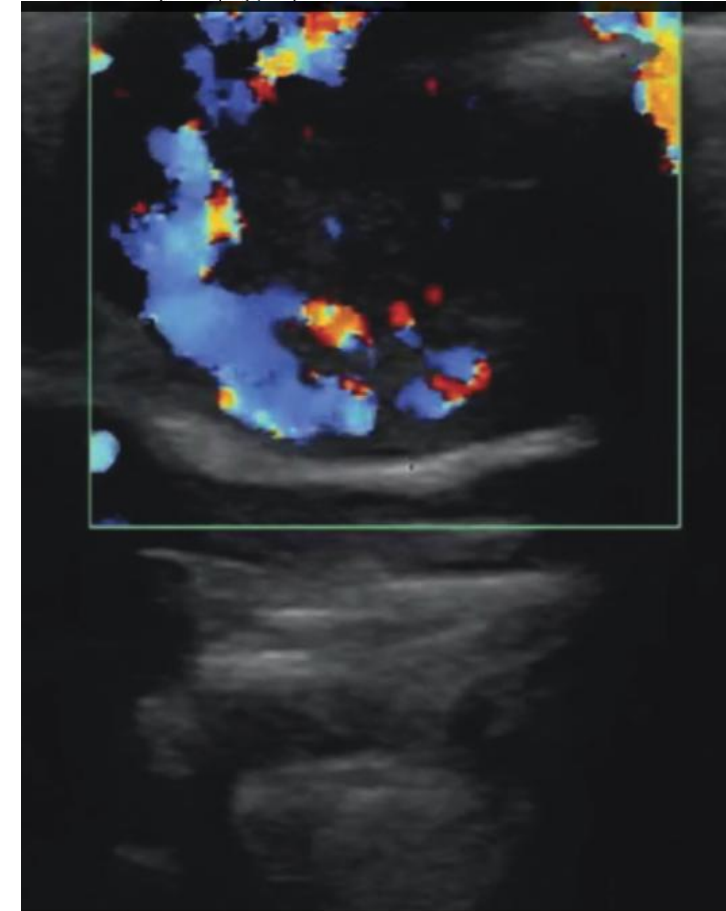
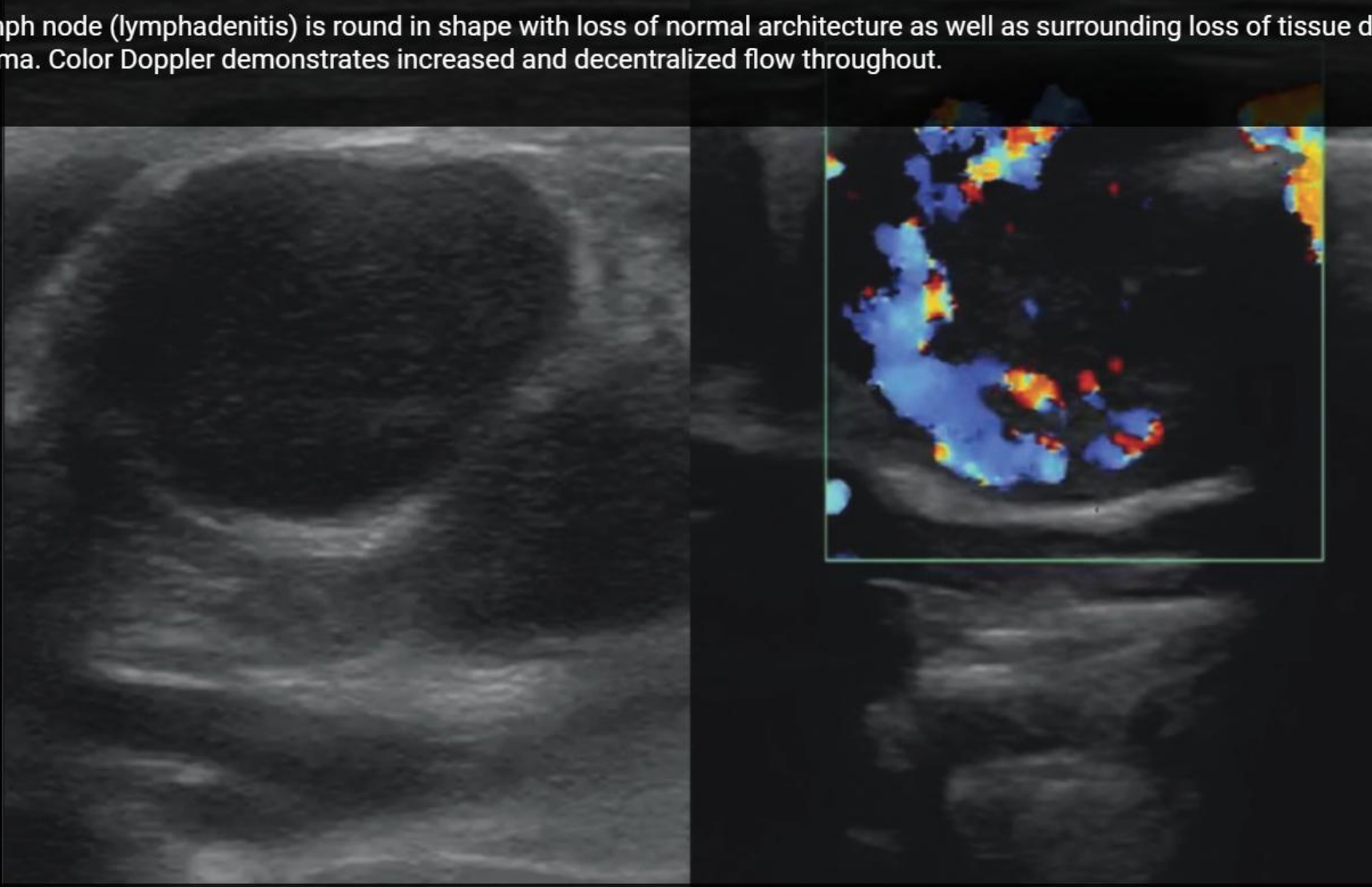


FIGURE 6-3. The infected lymph node (lymphadenitis) is round in shape with loss of normal architecture as well as surrounding loss of tissue definition, suggestive of early surrounding tissue edema. Color Doppler demonstrates increased and decentralized flow throughout.



Infected/Suppurative Lymph node

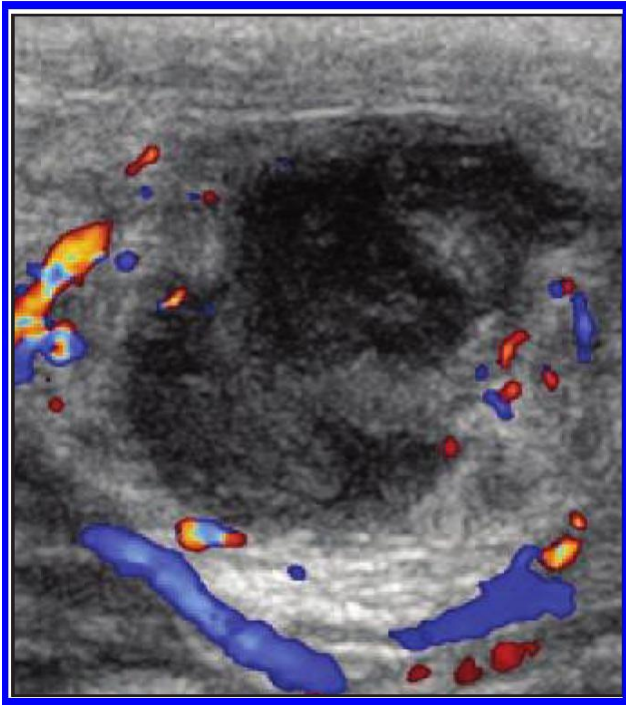
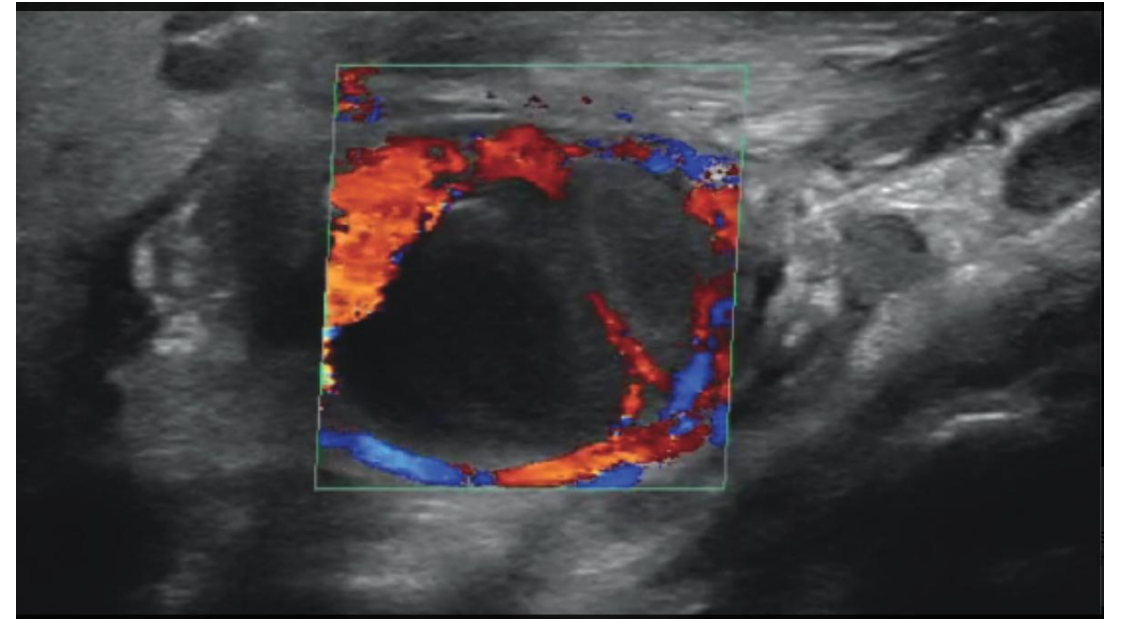
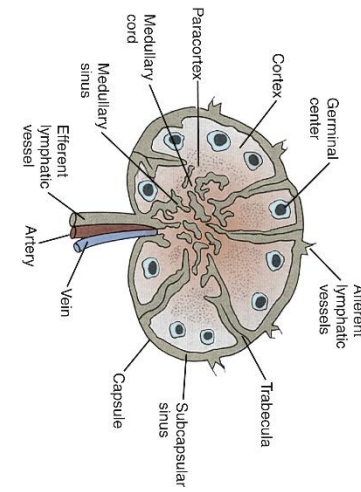
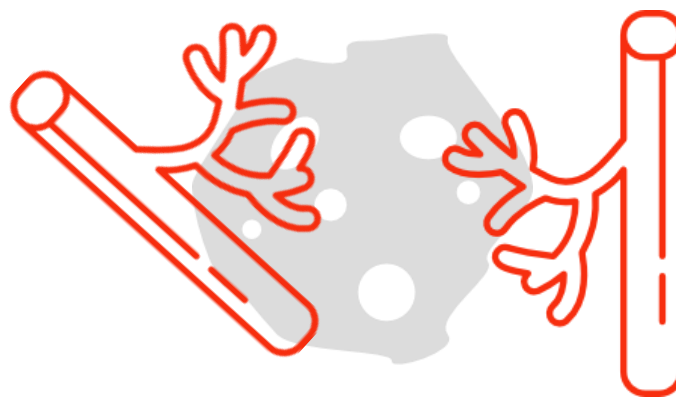


Fig. 5—6-year-old boy with suppurative lymph node due to *Staphylococcus aureus*. Gray-scale color Doppler ultrasound shows enlarged cervical lymph node with central hypoechoogenicity and peripheral Doppler vascularity.





- Ultrasonography findings that are concerning for malignancy include
- Increased node size (>0.5 cm)
- Variable echogenicity (can represent debris or hemorrhage)
- Presence of abnormal vasculature.



- Malignant features can also include
- Intranodal necrosis
- Calcification
- Ill-defined borders
- Matting
- Adjacent soft tissue edema

Malignancy

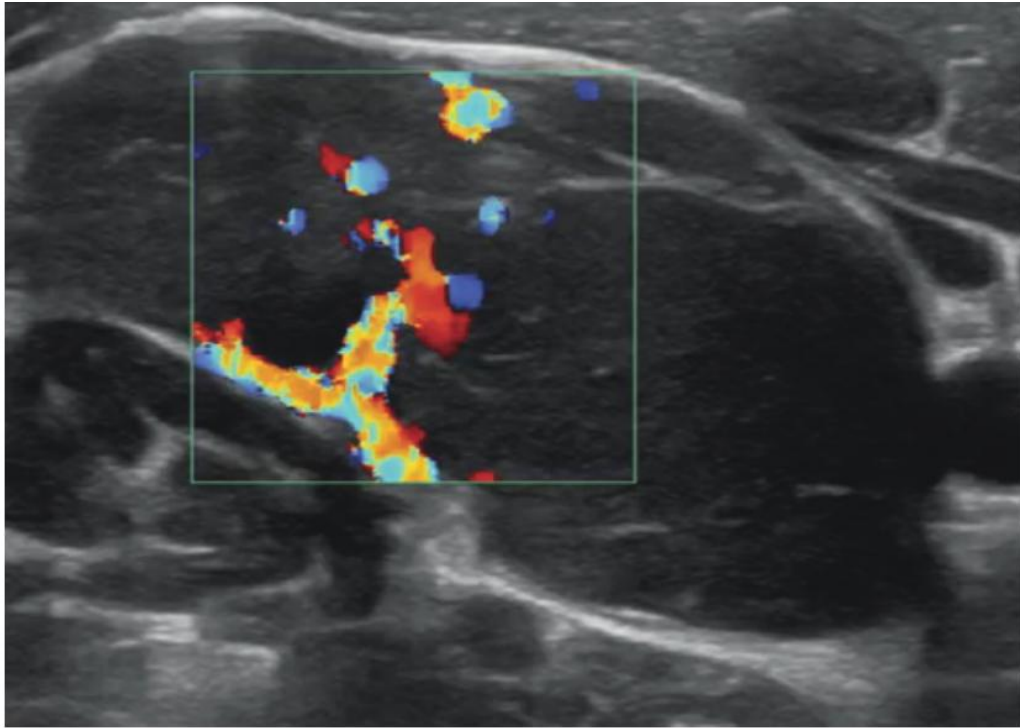
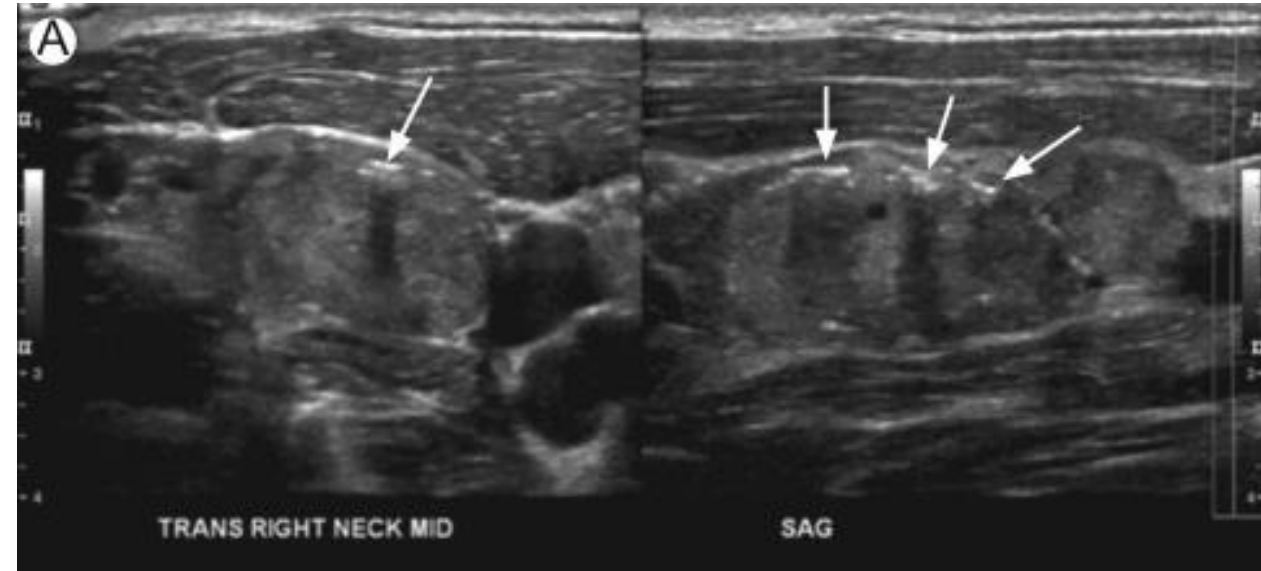


FIGURE 6-5. The malignant lymph nodes are enlarged, rounded, and lack normal lymph node architecture, with no defined hilum or cortex. Color Doppler reveals mixed vascularity rather than the typical central hilar flow on normal lymph nodes. This patient was diagnosed with lymphoma.



Progressive involvement of a benign lymph node with malignancy with progressive disappearance of the lymph node hilum.

